# LSTM Practice Question

Q1 Approx: $\tanh 4.0 = 1$

Soln Given: $C_{t-1} = 4$, $h_{t-1} = 0.5$, $f_t = 0.7$, $i_t = 0.5$, $\hat{C}_t = 2.4$, $o_t = 0.6$

- $C_t = (f_t \times C_{t-1}) * (i_t \times \hat{C}_t)$
  $= (0.7 \times 4) + (0.5 \times 2.4)$
  $= 4$
- $h_t = (o_t) \times \tanh(C_t)$
  $= 0.6 \times \tanh(4)$
  $= 0.6$

$C_t + h_t = 4 + 0.6 = 4.6$

updated cell state = 4.6

Q2 LSTM forward pass

Given: $x_t = 0.5$, $h_{t-1} = 0.2$, $C_{t-1} = 0.4$

- forget gate: $f_t = \sigma(w_f x_t + U_f h_{t-1} + b_f)$

$w_f = 0.7$, $U_f = 0.3$, $b_f = 0.1$

$f_t = \sigma((0.7)(0.5) + (0.3)(0.2) + 0.1)$
$= \sigma(0.51)$

$f_t = \sigma(0.51)$
$= \frac{1}{1+e^{-0.51}} = 0.624$

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- Input gate $i_t$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

Given: $W_i = 0.6$, $U_i = 0.4$, $b_i = 0$

$$i_t = \sigma((0.6)(0.5) + (0.4)(0.2) + 0)$$

$$i_t = \sigma(0.30 + 0.08)$$
$$i_t = \sigma(0.38)$$

$$i_t = \frac{1}{1+e^{-0.38}} = 0.593$$

- $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$

$W_c = 0.9$, $U_c = 0.2$, $b_c = 0$

$$\tilde{C}_t = \tanh[(0.9)(0.5) + (0.2)(0.2)]$$
$$= \tanh(0.49)$$

$$\tanh(0.49) = \frac{e^{0.49} - e^{-0.49}}{e^{0.49} + e^{-0.49}} = 0.454$$

- $C_t = f_t C_{t-1} + i_t \tilde{C}_t$

$$C_t = (0.624)(0.4) + (0.593)(0.454)$$
$$C_t = (0.249) + (0.269)$$
$$= 0.519$$

- $O_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$

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Given: $W_o = 0.5$, $U_o = 0.5$, $b_o = 0.1$

$$O_t = \sigma\left((0.5)(0.5) + (0.5)(0.2)\right)$$

$$= \sigma(0.45)$$

$$O_t = 0.61$$

- $h_t = O_t \tanh(C_t)$

$$= (0.61)(0.477)$$

$$= 0.2916$$

Q3 Given: Input Size = 10
Hidden Size = 32

(a) One Bias per gate

- For one gate: Input weights : $10 \times 32 = 320$
Hidden weights : $32 \times 32 = 1024$

Bias = 32

$\therefore$ Total = $320 + 1024 + 32 = 1376$

For four gates:

$4 \times 1376 = 5504$

(b) Two bias per gate

- Now Each gate Contains two bias vector

$32 + 32 = 64$

parameter per gate : $320 + 1024 + 64 = 1408$

- 4 gates: $4 \times 1408 = 5632$

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Q4 Vanishing gradient through Cell state

$f^k$, where $f$ = forget gate value
$k$ = no. of timestep

$f = 0.9$

After 10 steps: $f^{10} = (0.9)^{10}$
$= 0.3487$

After 50 steps: $f^{50} = (0.9)^{50}$
$= 0.0052$

(b) After 50 steps

Case 1: $f = 0.5 = (0.5)^{90}$
$= 8.8818 \times 10^{-16}$

Therefore: $f^{50} \approx 0.0000\ldots$

Thus the gradient has essentially disappeared.

Case 2: $f = 0.99$

$(0.99)^{50}$

$f^{0.99} = 0.605$

Thus approx 60% gradient remains.

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